Computer Skills 2

Functions Assignment

```
#include<iostream.h>

void duplicate ( int a[] )
{
for ( int i=0; i<10; i++)
{
a[i] = a[i] * 2;
}
}

void main ()
{
int z[10] = {1,2,3,4,5,6,7,8,9,10};
duplicate (z);
for(int i=0;i<10;i++)
cout<<z[i]<<"   ";
}
```

Output
2 4 6 8 10 12 14 16 1 8 20

```
#include<iostream.h>

int maximum ( int a[] )
{
int max = a[0];
for ( int i=0; i<10; i++)
{
if (a[i]>max)
max=a[i];
}
return max;
}

void main ()
{
int z[10] = {1,2,3,4,5,6,7,8,9,10};
cout<<maximum(z);
}
```

Output
10

`#include <iostream.h>`
 	
`int f1(int x, int num)`
 	
`{`
 	
`num= x++;`
 	
`return num+2;`
 	
`}`
 	
`void main()`
 	
`{`
 	
`int z=2;`
 	
`z=f1(z,3);`
 	
`cout<<z;`
 	
`}`
	Output
	4

`#include<iostream.h>`
 	
`int f1(int z)`
 	
`{`
 	
`return z*z;`
 	
`}`
 	
`void main()`
 	
`{`
 	
`int x=2;`
 	
`cout<<f1(f1(f1(x)));`
	Output
	256

}
 	
#include<iostream.h>
 	
void f1();
void f2();
void f3();
void f4();
 	
void main()
{
f1();
}
 	
void f1()
{
cout<<1;
f2();
cout<<1;
}
 	
void f2()
{
cout<<2;
f3();
cout<<2;
}
void f3()
{
cout<<3;
f4();
cout<<3;
}
void f4()
{
cout<<4;
}
	Output
	1234321



``
#include<iostream.h>
 	
void func(int &x, int y)
 	
{
 	
x=33;
 	
y=4;
 	
cout<<x<<" "<<y<<endl;
 	
}
 	
void main()
 	
{
 	
int a=4;
 	
int b=9;
 	
cout<<a<<" "<<b<<endl;
 	
func(a,b) ;
 	
cout<<a<<" "<<b<<endl;
 	
}
``
	Output
	4 9
	33 4
	33 9



```
#include<iostream.h>

    int x;
    void m1(int);
    int m2(int);

    void main()
    {
     int x=10;
     x+=2;
     m1(x);
     cout<<m2(5)<<endl;
    }

    void m1(int a)
    {
     x*=10;
     x=m2(a);
     cout<<"x="<<x<<endl<<"a="<<a<<endl;
    }

    int m2(int b)
    {
    return b*2;
    }
```

Output
x=24
a=12
10

```
#include<iostream.h>
int test (bool x, int y);
int z=10;

void main()
{    int b=20;
     cout<<test(false,b)<<endl;
     cout<<test(b,b)<<endl;
}

int test(bool a, int b)
{    int t;
     if(a)
     {
          t=b+z;
          return t;
     }
     else
     {
          t=z-b;
          return t;
     }}
```

Output
-10
30

include<iostream.h>
increment1();
increment2();
void main()
{
int c;
c=increment1(); cout<<c<<endl;
c=increment1(); cout<<c<<endl;
c=increment2(); cout<<c<<endl;
c=increment2(); cout<<c<<endl;
}
 	
increment1()
{
static int n=10;
n=n+2;
return n;
}
 	
increment2()
{
int n=10;
n=n+2;
return n;
}
	Output
	12
	14
	12
	12

#include<iostream.h>
 	
int test (int i)
{
static int x[]={1,2,3,4};
x[i]++;
return x[i];
}
 	
void main()
{
cout<<test(0)<<endl;
cout<<test(0)<<endl;
}
	Output
	2
	3

```
#include<iostream.h>
void m1(int, int&);

void main()
{
     int x=10, y=20;
     m1(x,y);
     cout<<x<<endl<<y<<endl;
}

void m1(int a, int& b)
{
     a+=2;
     b+=2;
     cout<<a<<endl<<b<<endl;
}
```

Output

12
22
10
22

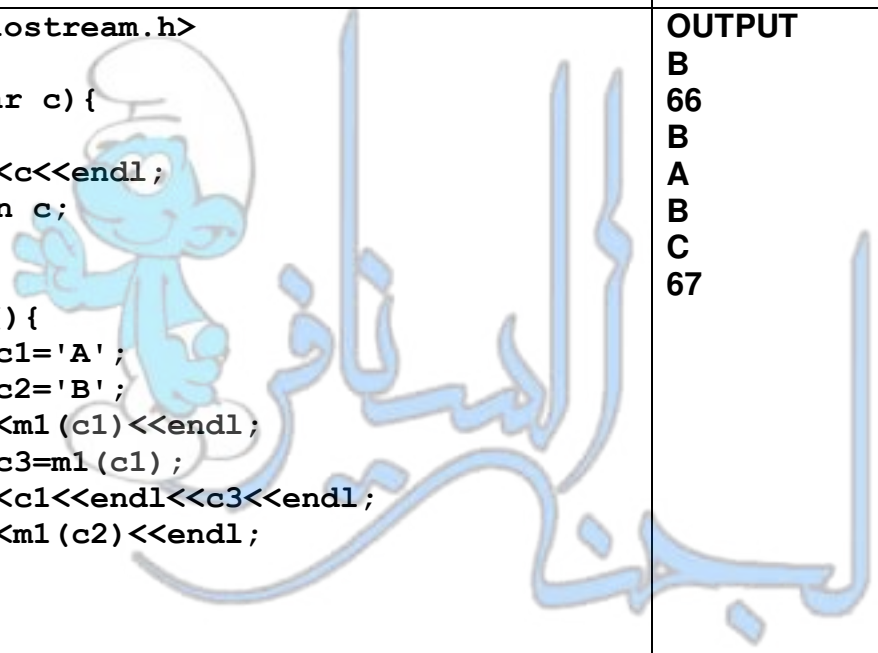
```
#include<iostream.h>

int m1(char c){
     c++;
     cout<<c<<endl;
     return c;
}

void main(){
     char c1='A';
     char c2='B';
     cout<<m1(c1)<<endl;
     char c3=m1(c1);
     cout<<c1<<endl<<c3<<endl;
     cout<<m1(c2)<<endl;
}
```

OUTPUT

B
66
B
A
B
C
67



```
#include<iostream.h>
#include<math.h>

bool m1(float);
void m2(int);

void main()
{
     int x=10;
     float y=10.2;
     if(m1(y))
          cout<<"y is integer"<<endl;
     else
          cout<<"y is not integer"<<endl;
     m2(x);
}

bool m1(float f)
{
     if(ceil(f)==f)
          return true;
     else
          return false;
}

void m2(int a)
{
     for(int i=1;i<=a;i++)
          cout<<"*";
     cout<<endl;
}
```

Output

```
y is not integer
**********
```

```
#include<iostream.h>

m1(int x, int y){
     y++;
     x+=y+5;
     cout<<y<<endl<<x<<endl;
     return 10;
}

void main(){
     int a=5, b=3;
     cout<<m1(a,b)<<endl;
     cout<<m1(b,a)<<endl;
     m1(20,5);
}
```

Output

```
4
14
10
6
14
10
6
31
```

`#include<iostream.h>`
 	
`int max(int n1, int n2){`
`  if(n1>n2)`
`  return n1;`
`  return n2;`
`}`
 	
`void main(){`
`  int x=10, y=2, z=50;`
`  cout<<max(x,max(y,z))<<endl;`
`}`
	Output
	50

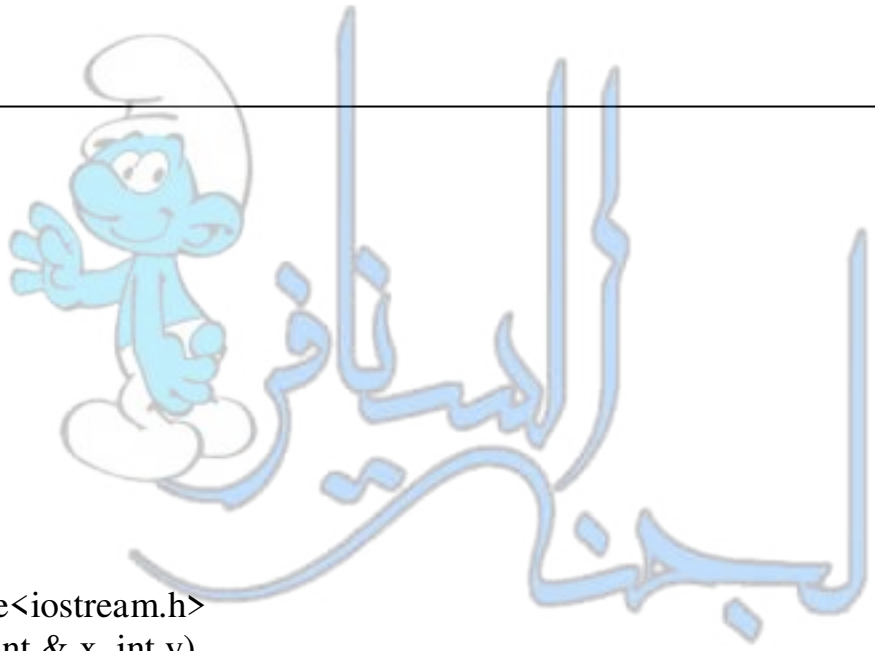
`#include<iostream.h>`
 	
`int m1(int n1, int n2)`
`{`
 	
`  return 12;`
`  cout<<"**********"<<endl;`
`  return n1+n2;`
 	
`}`
 	
`void main()`
`{`
 	
`  cout<<m1(2,3)<<endl;`
 	
`}`
	Output
	12

```
void main()
{
        int n= 0;
          for (int h=0; h < 10; h++)
        {
           if ((h % 3)==0 || (h % 4)==0)
              continue;

           cout << h <<"  ";
        }
```

```
#include<iostream.h>
int test ( int a, int b, char z)
{
a=7;
cout<<++z;
return a*b;
}
void main()
{
cout<<test(4,2,'J');
}
```

Output:
K14



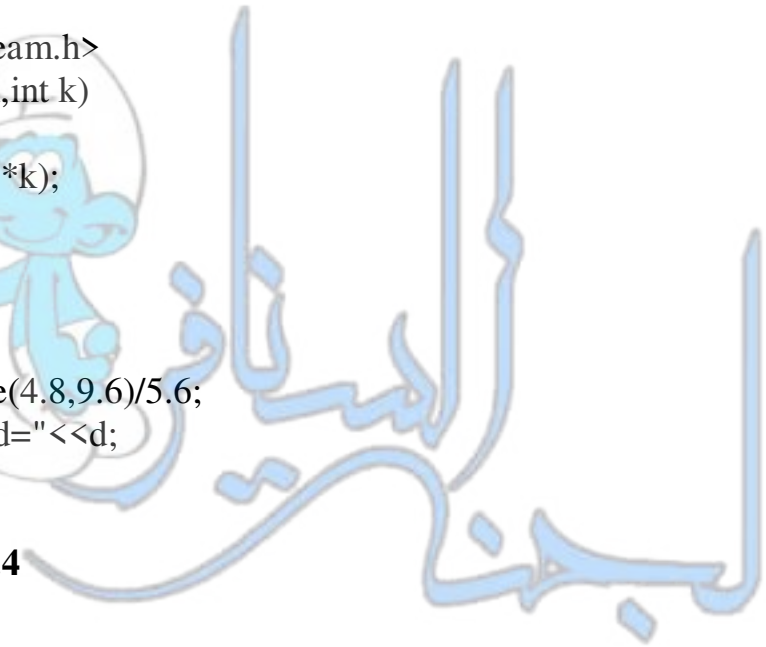
```
#include<iostream.h>
int fun(int & x, int y)
{
x--;
y = y/2;
cout << "x=" << x << " y=" << y << "\n";
return y - x;
}

void main()
{
int a = 3;
int b = 6;
```

```
while ( b>0 )
{
b = fun(a,b);
cout << "a=" << a << " b=" << b << "\n";
}
}
```

Output:
x=2 y=3
a=2 b=1
x=1 y=0
a=1 b= -1

```
#include<iostream.h>
int square(int h,int k)
{
        return (h*k);
}
void main()
{
        float d;
        d=square(4.8,9.6)/5.6;
        cout<<"d="<<d;
}
```



Output: d=6.4